

INTRODUCTION

The horse **raced** past the barn fell.

- Garden path sentences (like the one above) are difficult to process because **they are temporarily ambiguous**
- *Raced* is probably assumed to be the action done by the horse until we reach *fell*... which has no agent! Somehow this sentence will have to be reanalyzed to reach the correct understanding
- The difficulties in garden path processing arise because of the **sequential nature of human language processing**... when we see *raced*, we don't know that *fell* will still come later
- However, other language systems (like BERT) process their input all at once. Does this mean they could be better than humans at garden path processing?

Garden path sentences, like “The Horse Raced Past the Barn Fell”, contain temporary ambiguity as to the correct syntactic parse of the sentence. In our example sentence, the horse is assumed to be the agent of “raced”, until we arrive at the disambiguating information, in this case, the verb “fell”.

Human readers tend to comprehend garden path sentences poorly. Explanations for this failure to comprehend are varied. On the one hand, it could be that readers fail to construct the correct syntactic structure of the garden path sentence, even after receiving the disambiguating information. Another explanation is that while readers do achieve the correct syntactic parse, the roles assigned to the initial parse stick around, causing interference in comprehension. These struggles highlight the largely linear nature of human sentence processing. However, other language systems, like “BERT”, process their input alinearly. This leads to our study’s main questions (next slide)

OBJECTIVES

Question 1: Given that BERT and humans process language differently, do the two also differ in the extent to which they can extract information from garden path sentences in order to correctly answer questions about them?

Question 2: Humans might have difficulty processing garden path sentences because of semantic role misassignment. Is this also a challenge that BERT faces when encountering garden paths?

First, (READ QUESTION 1). Additionally, since humans may have difficulty processing garden path sentences due to semantic role mis-assignment, is this mis-assignment also present in BERT's embeddings?

QUESTION ANSWERING

40x4 Sentences [3]	Garden Path	The child _{1,3} (who was) bought ₅ the ice cream cone hugged ₄ the nanny ₂
	Local Coherence	The nanny ₁ hugged ₄ the child _{2,3} (who was) bought ₅ the ice cream cone
5x QuestionTypes	¹Agent Matrix	Q: Who gave a hug? A: {The child/the nanny}
	²Patient Matrix	Q: Who got a hug? A: {The nanny/the child}
	³Ambiguous Arg.	Q: Who got an ice cream cone? A: The child
	⁴Matrix Action	Q: What did {the child/the nanny} do? A: Hugged
	⁵Embedded Act.	Q: What did the child get? A: Bought the ice cream cone

The first step to addressing these questions was creating a set of garden path sentences and questions about them that we could give to humans and a BERT question answering model to see how the two differed in performance.

We adapted our garden path sentences from stimuli used by Kiel Christianson in a previous experiment. In addition to garden path sentences, he also included another structure called local coherence sentences. These are very similar to garden path sentences, except instead of the relative clause coming after the first noun phrase (where it could be misinterpreted as the main

verb of the sentence), it comes after the second sentence. In this position, it should not be ambiguous because the main verb has already been encountered. We included these in our stimuli set as a comparison of how difficult sentences with relative clauses are to process even if they do not lead to temporary ambiguity.

A second manipulation we included was the transparency of the relative clause. We can disambiguate “the child bought the ice cream cone hugged the nanny” simply by adding “who was” at the beginning of the relative clause to say “the child who was bought the ice cream cone hugged the nanny”. This disambiguates the relative clause in the garden path sentence and marks it further in the local coherence structure, so we included this manipulation as well to see how much humans and models are able to benefit from an explicit disambiguation cue.

We also created a set of question items to probe the different relationships between words in the sentences. The first set provide a verb in the question and the answer should be an entity from the sentence. In the first type (Agent Matrix), we aim to see if the reader makes a connection between the main verb of the sentence and its agent. So for example in these sentences, hug is the main verb, and the answer to “who gave a hug?” would be “the child” for the garden path sentence, and “the nanny” for the local coherence sentence.

The second type (Patient Matrix) asks who receives the action of the main verb. So “who got a hug?” would be “the nanny” for the garden path structure, and “the child” for the local coherence sentence.

The third type (Ambiguous Argument), probes the relationship between the verb in the embedded clause and the noun phrase which the clause modifies. As mentioned previously, it is not immediately clear that the embedded clause is modifying rather than the main verb when read linearly, so that might lead readers to the incorrect interpretation that the noun is doing the action in the embedded clause rather than receiving it. To test this we asked, for example, “who got an ice cream cone?” to which the answer would be “the child” for both sentence types. We also asked two additional questions about the actions done by a particular entity. They probe information similar to the questions already described, but from a different perspective. In the first (Matrix Action) we ask what the first noun phrase entity does, and the answer should be the matrix verb (“hugged” for both garden path and local coherence sentences presented here). In the final question, we ask what the entity modified by the relative clause received, and the answer should be “bought an ice cream cone” for both structures presented here.

In total we had 40 sets of stimuli, with each set made of 1 ambiguous and 1 unambiguous garden path and 1 ambiguous and 1 unambiguous local coherence sentence for a total of 160 sentences. For each of these we also had 5 question types, which gave us 800 context-question pairs in total.

Behavioral Experiment

- 74 English speakers recruited from Prolific
- 45 random items: 32 target + 13 filler
- > 80 responses for each of the sentence and question types

BERT Experiment

- Same 800 context + question pairs
- Pre-trained BERT_{BASE} model which had been fine-tuned for QA using SQuAD2.0

The nanny hugged the child
bought the ice cream cone.

Who hugged the child?

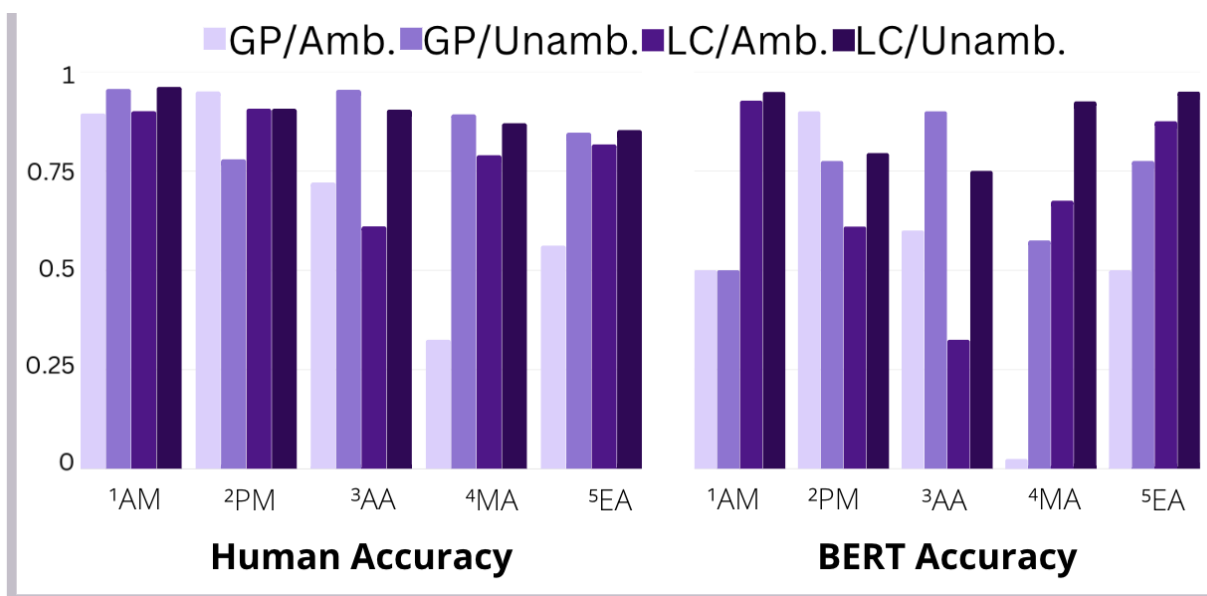


Context: The nanny hugged the
child bought the ice cream cone.
Question: Who hugged the
child?



We ran two experiments to compare the ability of BERT and humans to extract information from garden path sentences. In the human experiment, this was done by presenting the full sentence to the reader, then when they indicated they were finished reading, presenting a question which they could respond freely to. We had 74 participants, and each participant saw a random set of 45 total items, containing a mixture of items from our stimuli set and filler items. This meant that there were at least 80 separate items for each question type of interest.

For the BERT portion of the experiment, we used the same 800 context and question items. We provided them both to a pretrained BERT_{base} model which had been previously fine tuned to perform question answering tasks using the SQuAD2.0 dataset. We recorded each of the answers from humans and BERT, scored them, and then analyzed the average accuracy for question types and sentence types.



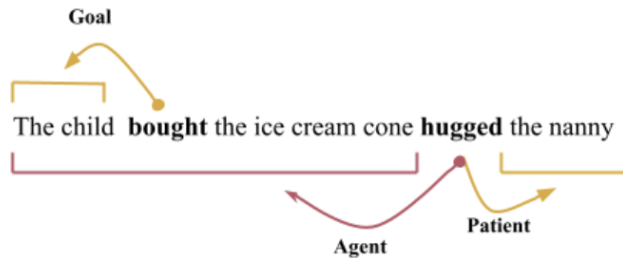
As you can see in the graph on the left, humans perform very well for most of the sentence types. Performance is worst on ambiguous garden path sentences with matrix action questions, and it is also low for the same sentences with ambiguous argument questions and embedded action questions. The local coherence sentences generally have performance at least as good as the garden path sentences which supports the idea that they are easier to process. Additionally, unambiguous sentences have generally higher accuracy than their corresponding ambiguous sentences, as expected.

BERT, on the other hand, does not perform as well as humans in extracting information from garden path sentences as can be seen in the graph on the left. This is especially true for the agent matrix and matrix action question types, which are the most difficult to answer. Local coherence structures had generally better scores, but in some cases they were still well below what we saw from the human performance. Additionally, unambiguous garden path sentences did not perform better than ambiguous versions for agent matrix and patient matrix questions. Both of these facts suggest that BERT has difficulty extracted information from sentences with relative clauses, and this contributes to relatively low accuracy in answering questions about garden path sentences.

PROBE TASK

Probe Training

We trained two binary linear classifiers on BERT's word embeddings (8th layer, following findings from [6]) from sentences annotated in PropBank [5] for semantic roles Agent and Goal.



Classifier	Accuracy	Selectivity
Agent Classifier	85	35
Goal Classifier	85	37

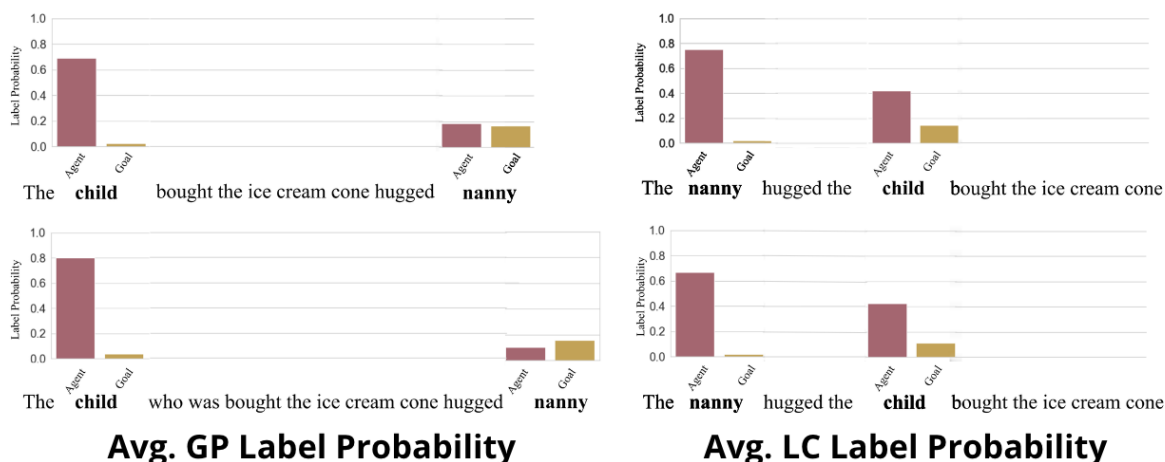
- 5000 training examples with a 50/50 split between having or not having the role in question.
- Chose hyperparameters to maximize selectivity [9]
- 80/20 train test split
- High accuracy on Propbank test sentences

Since BERT appears to perform poorly in comparison to humans when processing garden path sentences, we wanted to investigate the cause of these errors. Since humans may process garden path sentences poorly due to semantic role mis-assignment, we decided to investigate the roles BERT assigns to the different entities in garden path sentences.

In sentences like "The child bought the ice cream cone hugged the nanny", the "child" is both the agent of the verb hugged, and the goal of the verb bought. To correctly represent sentences like these, both roles need to be represented in some way on the token "child". In order to investigate cases in which multiple roles are assigned to one token (for example, "child"), we trained two binary linear classifiers to predict the semantic role based on BERT's word embeddings. We specifically chose the Agent and Goal roles for our classifiers. After annotating our stimuli sentences using PropBank annotation guidelines, the Goal role was the most common role for the first entity (e.g. the child) in our garden path sentences, along with the Agent role. The "Goal" role was sparsely represented in the PropBank corpus, and so our training sets contained a 50/50 split of tokens that did or didn't have the role in question. We then used an 80/20 train/test split, and achieved an accuracy of 85% for both the Agent and Goal classifiers.

Probe Testing on Garden Paths

- Tested on subset of context sentences from the question answering section, selecting only sentences containing both Agent and Goal roles, according to PropBank annotations (97/160 sentences total)



- BERT fails to assign the Goal role to the head of the embedded clause
- Nouns immediately preceding verbs are most likely to be agents regardless of the syntactic parse of the structure

After training, we applied the probe to the token embeddings of the first and second entities in our sentences, like “child” and “nanny”. We tested on only a subset of our sentences from the question answering portion, as we wanted to only target sentences that contained the goal role specifically. Here, we plot the average classifier probabilities for the first and second entities (seen in bold) of our sentences. For the first entity in garden path sentences (e.g. child), we see a high “agent” probability for both the ambiguous and unambiguous sentences. However, the average ‘goal’ probabilities are extremely low, with 2% for the ambiguous condition, and 4% for the unambiguous condition. For our local coherence sentences, such as “the nanny hugged the child bought the ice cream cone”, we see a high agent probability on “child”, despite the child not carrying an agent role at all in the correct parse of the sentence.

Assuming that our classifier is indeed decoding semantic role information, this indicates that BERT assigns semantic roles somewhat based on local context, relying on the typical agent-verb order, rather than utilizing longer-distance information that reveals the appropriate correct.

CONCLUSION

- **BERT is worse than humans at answering questions about garden path sentences** despite not being subject to the ambiguity
- It also shows difficulty in some cases with no ambiguity, suggesting **difficulty processing relative clauses** as well
- These results may be due to a **failure to assign the correct semantic role** in relative clauses, following theories from human sentence processing



PDF and Code

Overall, it appears that BERT does not meet human performance when it comes to question answering based on garden path sentences. This is perhaps somewhat surprising because BERT is not subject to the temporary ambiguity that leads to processing difficulty for humans due to the linear nature of language input.

Rather, BERT's difficulties may be due to difficulty processing relative clauses in general and its failure to assign the correct semantic roles, relying on heuristics based in typical agent-verb word ordering instead of long distance relationships between words.